

Laboratorio #6.1: Docker Compose + PostgreSQL con Flask (secrets, redes y volúmenes)

Laboratorio:

Objetivos:

- Corregir las tres debilidades identificadas en laboratorio_docker_compose_00.md
- Gestionar credenciales con secrets y env_file en vez de environment
- Aislar PostgreSQL con una red interna: true
- Persistir datos con un named volume
- Verificar cada corrección con comandos (docker inspect, network inspect, git check-ignore), no darla por hecha
- Conectar Flask a PostgreSQL con psycopg y connection pooling

Guías de laboratorio:

- https://github.com/wils0n/laboratorios-md/blob/main/laboratorio_docker_compose/laboratorio_docker_compose_01.md
- https://github.com/wils0n/laboratorios-md/blob/main/laboratorio_docker_compose/laboratorio_docker_compose_00.md (prerrequisito: lab 00, stack base con los riesgos identificados)

Entregables

1. **URL de repositorio** (GitHub o GitLab).
 - o <https://github.com/aarontimana04/Lab06-1.git>
2. Subir los cambios a la rama feature/flask-docker-compose a su repositorio usando buenas prácticas de git (GitFlow/TBD, conventional commits)

Estructura del repo:

Ejemplo:

```
flask-docker-app/  
├─ app.py  
├─ requirements.txt  
├─ Dockerfile  
├─ .dockerignore  
├─ .gitignore  
├─ .env.dev          (ignorado por git y por Docker)  
├─ pg_password.txt   (ignorado por git y por Docker)
```

└─ init.sql
└─ compose.yaml

3. Captura de su imagen multicontenedor corriendo

```
[+] Running 5/5
✔ Network lab06-1_public          Created      0.0s
✔ Network lab06-1_private         Created      0.0s
✔ Volume "lab06-1_postgres_data"  Created      0.0s
✔ Container lab06-1-postgres-1    Healthy      6.1s
✔ Container lab06-1-flask-1       Started      5.9s

[usuario@MacBookPro Lab06-1 % docker compose ps
NAME                IMAGE              COMMAND                  SERVICE    CREATED        STATUS               PORTS
lab06-1-flask-1     lab06-1-flask     "gunicorn --bind 0.0..." flask       3 minutes ago  Up 3 minutes        0.0.0.0:8080->5000/tcp
lab06-1-postgres-1  postgres:16.3     "docker-entrypoint.s..." postgres   3 minutes ago  Up 3 minutes (healthy)

[usuario@MacBookPro Lab06-1 % curl localhost:8080/
{"message":"Flask + PostgreSQL funcionando"}
[usuario@MacBookPro Lab06-1 % curl localhost:8080/items
[{"id":1,"priority":"high","task":"doctor appointment"},{"id":2,"priority":"low","task":"buy dinner"},{"id":3,"priority":"high","task":"lomo saltado"}]
usuario@MacBookPro Lab06-1 %
```

☐			lab06-1	-	-	-	0.04%			
☐			postgres-1	303a3d7bbbd4		postgres:16	0.01%			
☐			flask-1	ee44fafa1c14	lab06-1-fla	8080:5000	0.03%			

4. Captura que evidencie persistencia de datos: respuesta de GET /items después de docker compose down y docker compose up -d (paso 11).

```
[usuario@MacBookPro Lab06-1 % docker compose port postgres 5432
no port 5432/tcp for container lab06-1-postgres-1:
[usuario@MacBookPro Lab06-1 % curl localhost:8080/items
[{"id":1,"priority":"high","task":"doctor appointment"},{"id":2,"priority":"low","task":"buy dinner"},{"id":3,"priority":"high","task":"lomo saltado"}]
usuario@MacBookPro Lab06-1 %
```

```

[usuario@MacBookPro Lab06-1 % docker compose down
[+] Running 4/4
  ✓ Container lab06-1-flask-1      Removed      0.3s
  ✓ Container lab06-1-postgres-1   Removed      0.1s
  ✓ Network lab06-1_public         Removed      0.1s
  ✓ Network lab06-1_private        Removed      0.2s
[usuario@MacBookPro Lab06-1 % docker volume ls
DRIVER      VOLUME NAME
local       2f935e60bdea235ff229018f5aa7aa8598273d14e82e684ab44958217df3fade
local       3fad330cdaa4bac2674669b56c005f66f4fc42df48e54303aca83c0e627dca46
local       4c865766cda2c45f0b3875223ae2886953f94407a11db268a46fe88cccef3b03
local       7cd0e0083680b5c3cdc2eb96b9af71594d74944a9d0c61ef30dd4a6692358bd3
local       c2e3316c0c325d6d0a0ab7c1e01498c5f8978b105e5c8b08730d91e00e61cdfc
local       d9837a0699f28b566a0308ac81d2ce7a818b77c56ba7f6742171bd866e137316
local       ec5039fb5b33c7da9424748ac4471d44dff523180696c512ca47be0163041444
local       fd157fd9bbd7c7f4c18dbec90e6f0897bfc43289f227a7e0d6282c14aabeeb0ff
local       lab06-1_postgres_data
local       snipe-it_db_data
local       snipe-it_storage
[usuario@MacBookPro Lab06-1 % docker compose up -d
[+] Running 4/4
  ✓ Network lab06-1_public         Created      0.0s
  ✓ Network lab06-1_private        Created      0.0s
  ✓ Container lab06-1-postgres-1   Healthy      5.8s
  ✓ Container lab06-1-flask-1      Started      5.8s
[usuario@MacBookPro Lab06-1 % docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED   STATUS    PORTS
lab06-1-flask-1     lab06-1-flask       "unicorn --bind 0.0..." flask        8 seconds ago Up 3 seconds 0.0.0.0:8080->5000/tcp
lab06-1-postgres-1 postgres:16.3        "docker-entrypoint.s..." postgres    8 seconds ago Up 8 seconds (healthy)
[usuario@MacBookPro Lab06-1 % curl localhost:8080/items
[{"id":1,"priority":"high","task":"doctor appointment"}, {"id":2,"priority":"low","task":"buy dinner"}, {"id":3,"priority":"high","task":"lomo saltado"}]
[usuario@MacBookPro Lab06-1 % ]

```

5. Captura de la verificación de credenciales (pasos 10.1 y 10.2): docker inspect <postgres> --format '{{.Config.Env}}' sin la contraseña en claro

antes:

```

[usuario@MacBookPro Lab06-1 % docker inspect lab06-1-flask-1 --format '{{.Config.Env}}'
[POSTGRES_USER=myuser DB_HOST=postgres DB_PORT=5432 DB_NAME=mydb DB_USER=myuser POSTGRES_DB=mydb PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin LANG=C.UTF-8 GPG_KEY=7169605F62C751356D054A26A821E680E5FA6305 PYTHON_VERSION=3.12.14 PYTHON_SHA256=5c8462af5790baf43a321a1559dbe0db06d1be4300fb85fb53c40060668e548a]
[usuario@MacBookPro Lab06-1 % ]

```

después:

```

● (base) → flask-docker-app git:(main) x docker inspect flask-docker-app-postgres-1
--format '{{.Config.Env}}'
[POSTGRES_DB=mydb POSTGRES_PASSWORD_FILE=/run/secrets/pg_password POSTGRES_USER=myser
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/lib/postgresql/16/bin
GOSU_VERSION=1.17 LANG=en_US.utf8 PG_MAJOR=16 PG_VERSION=16.3-1.pgdg
20+1 PGDATA=/var/lib/postgresql/data]
○ (base) → flask-docker-app git:(main) x

```

Nota: Tener cuidado de no trackear en git .env.dev y el pg_password.txt

6. Captura de la verificación de red (pasos 10.4 y 10.5): docker network inspect <proyecto>_private --format 'internal={{.Internal}}' con internal=true, y docker compose port postgres 5432 sin salida.

```

573651422d33 snipe-it_default bridge local
[usuario@MacBookPro Lab06-1 % docker network inspect lab06-1_private --format 'internal={{.Internal}}'
internal=true

```

7. (Reto, ejercicio 12) Captura de la migración a Forma B: docker compose exec flask env | grep DB_PASSWORD sin resultado y GET /items respondiendo.

```

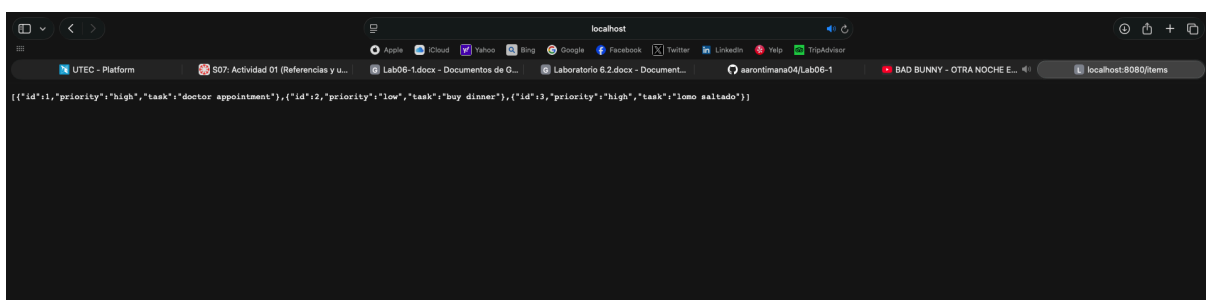
[usuario@MacBookPro Lab06-1 % docker compose exec flask env | grep DB_PASSWORD
]
[usuario@MacBookPro Lab06-1 % curl localhost:8080/items
[{"id":1,"priority":"high","task":"doctor appointment"}, {"id":2,"priority":"low","task":"buy dinner"}, {"id":3,"priority":"high",
"task":"lomo saltado"}]
]
[usuario@MacBookPro Lab06-1 % git status
On branch feature/flask-docker-compose

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .dockerignore
        .gitignore
        Dockerfile
        app.py
        compose.yaml
        init.sql
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
]
[usuario@MacBookPro Lab06-1 % git check-ignore -v .env.dev pg_password.txt
.gitignore:1:.env.dev      .env.dev
.gitignore:2:pg_password.txt  pg_password.txt
usuario@MacBookPro Lab06-1 %

```



The screenshot shows a web browser window with the address bar set to localhost. The page content displays the JSON output of a curl request to localhost:8080/items. The output is a list of three tasks, each with an id, priority, and task name.

```

[{"id":1,"priority":"high","task":"doctor appointment"}, {"id":2,"priority":"low","task":"buy dinner"}, {"id":3,"priority":"high","task":"lomo saltado"}]

```

Sorteo: 1 afortunado nos va a mostrar que lee de un solo archivo el password